Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/05/2026
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Single-Source Electromagnetic Levitation Flight Support for Ternary Alloy Rapid Solidification (ELFSTARS)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	04/26/2024	**End Date:**	04/25/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Hanson, Ian	**Contact Phone:**	
Contact Email:	ian.m.hanson@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per the PI: The Principal Investigator (PI) has changed his academic affiliation with Tufts University and now serves as a Full-time Research Professor in support of this grant. Consultant Dr. Jannatun Nawer has transitioned from Industry to a faculty position at Boise State University where she will continue her involvement in a new role: Co-Investigator and Institutional PI (Ed., 4/20/26).		
COI Name (Institution):	Nawer, Jannatun Ph.D. (Boise State University)		
Grant/Contract No.:	80NSSC24K1119		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This research addresses fundamental issues relating to modeling alloy behavior prediction, phase selection, and analysis of the processes governing microstructural evolution, which lead to improved modeling of Additive Manufacturing (AM) processes in space, low gravity, and on Earth.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Supports Integrated Computational Materials Engineering process optimization efforts on exploration-relevant aerospace alloy characterization.
Task Progress:	Batch 5 Ni-based ternary analog alloy samples and the appropriate documentation were prepared at the NASA Marshall Space Flight Center (MSFC) Levitation Lab and delivered to the European Space Agency (ESA) and Airbus for integration into the sample transfer mechanism for upload to the International Space Station (ISS) on HTV-X3 from the Tanegashima Space Center in Japan Spring/Summer 2027. Successfully conducted ISS space testing on Inconel 718, High Chromium Stainless Steel, High Nickel Stainless Steel, and elemental Gold.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Jeon S, Lee JJ, Kaban I, Matson DM. "In-situ observation of a transient intermediate state during metastable solidification of a Fe–Cr–Ni alloy." Acta Materialia. 2026 Feb 15:122028. https://doi.org/10.1016/j.actamat.2026.122028 , Feb-2026
Articles in Peer-reviewed Journals	Matson DM, Nawer J, Bergeron K, Phillips B. "Expressing uncertainty using regression and GUM for derived property measurements based on uncertainty in input quantities." High Temperatures-High Pressures. 2025 Nov 15;54(6):533-55. https://www.oldcitypublishing.com/wp-content/uploads/2026/04/HTHPv54n6p533-555.pdf , Nov-2025
Awards	Moldovean N. "Tufts University Mechanical Engineering Best Bachelors Honors Thesis Award, May 2025." May-2025
Dissertations and Theses	Muusha P. "Thermophysical property measurement and modelling of Ni-Cr-(Fe, Mo) superalloy analogs." PhD Dissertation, Tufts University Mechanical Engineering, February 2026. , Feb-2026
Dissertations and Theses	Moldovean N. "It is rocket science: System-wide sounding rocket optimization via design, requirements management, and testing." BS Honors Thesis, Tufts University Mechanical Engineering, May 2025. , May-2025
Dissertations and Theses	Fu B. "Thermodynamic modeling of the evaporation of ternary analog alloys of Inconel 625 and Inconel 718 in electrostatic levitation." BS Honors Thesis, Tufts University Physics Department, May 2025. , May-2025
Journal/Magazine covers	Irace PH, Dang C, Dutta P, Luo T, Matson DM, Michael F, Nawer J, O'Connor A, Rajagopalan G, Reeves RD, Roberts MS, Sansoucie MP, Weber R. "Cover illustration for NASA publication, 'A Researcher's Guide to: Microgravity Materials Research'." International Space Station (ISS) Researcher's Guide. Published by the NASA ISS Program Research Integration Office and written in collaboration with the ISS National Laboratory, which is managed by the Center for the Advancement of Science in Space (CASIS). Executive Editor J. Hamilton. https://www.nasa.gov/wp-content/uploads/2025/06/np-2025-jsc-microgravity-materials-iss-mini-book-final.pdf?emrc=5cbd59 , May-2025